# 2018

# CALENDAR YEAR REPORT

# WASTEWATER TREATMENT FACILITIES

## City of Memphis

## Division of Public Works

DONALD J. HUDGINS
REGISTERED ENGINEER
AGRICULTURE
COMMERCE
No. 23401
STATE OF TENNESSEE

Donald J. Hudgins

10-31-2019

Prepared by Don Hudgins, P.E.

Prepared on October 31, 2019

2018 Calendar Year Report
Wastewater Treatment Facilities
City of Memphis

## 1.0 Maxson Wastewater Treatment Facility

Introduction-

The Maxson Facility was plagued with a variety of major equipment breakdowns throughout the year resulting in various permit violations. Secondary Clarifier 3N failed on 1-23-2018 as well as a pipe gallery lift station. Raw sewage pumping was hampered by a bent shaft in RSP#1 in January and problems continued throughout the year with the new Pump Building IPA not providing sufficient cooling for VFDs. Elevated Mississippi River stages (39.4 feet) resulted in high influent flow in March causing additional violations.

About 4-10-2018, City officials notified that Horn Lake Basin Authority that the sewer agreement expiring 9-22-2023 would not be renewed. This would potentially reduce the Maxson Facility influent flow by 10-12 mgd.

The Maxson Facility was plagued with ruptures of the 18-inch primary sludge in July and August resulting in permit violations. There had been a total of seven such ruptures since 2015 of increasing severity. The pipeline was previously used for waste activated sludge until primary sludge wasting began in 2004. The pipeline likely experienced accelerated wear and tear due to the higher grit and sand content of primary sludge. Scott Contractors completed an emergency temporary repair using 16 joints of HDPE by mid-October and Street Maintenance re-surfaced the area with new asphalt. An engineering firm was engaged toward the end of the year to design and bid out a replacement pipe.

More emphasis was made on grit removal subsequent to the primary wasting line failure. Around mid-August, two grit chamber drain valves were discovered open and clogged. Once the drains were cleared and the valves closed, influent flow decreased by about 5 mgd and normal grit removal operations resumed. Severe effluent channel erosion beginning with the use of the Effluent Bypass Outfall channel was observed on 10-30-2018 and subsequently repaired by 11-7-2018 by Heavy Equipment using 24 + loads of stone.

As in years past, the Maxson Facility operated with four 1500 hp aeration blowers in Trickling Filter mode with at least one backup. This led to the presumption that a minimum of 4 duty blowers were required to operate the plant and one backup available. In actuality, it is better to have 5 duty blowers as required by ABF Mode and one backup available so that transition to this mode of operation will be seamless. There were 4 blowers available in July and by the end of November, only 3 were available resulting in deteriorating treatment performance due to motor bearing failure causing in permit violations. This resulted in the purchase of a refurbished blower motor and another new blower motor the following year as a shelf spare.

The City submitted a letter of interest in mid-2018 for WIFIA funding for Lagoon 5 expansion, primary sludge pumping improvements, new admin building, Lagoon 2A remediation, and new belt filter presses for secondary dewatering representing approximately $144 million in project value. EPA announced on 11-1-2018 that the projects would be funded pending submittal of a loan application by 7-6-2019. WIFIA would fund 49% of $144 million in project value. The City would need to come up with the remaining money.

CDM Smith began excavation of the new disinfection tank for GMP #1 on 7-2-2018. Effluent bypass using the effluent pumping station began on 10-3-2018 with the two chlorine contact tanks as wet wells. One of the pumps blew a fuse on 10-5-2018, flooding the disinfection tank excavation. As a result, the effluent pumps were put on 24/7 watch for the remainder of the project (next 12 to 18 months.) CDM Smith began pouring concrete for new disinfection tank on 12-24-2018.

The Maintenance Department performed numerous noteworthy tasks during the year. In January, staff made alterations to record storage room turning it into a fitness room, painted walls and ceiling and put matting on floor. Work out machines and other fitness equipment was installed to help employees stay healthy. The failed motor drive of ABF Pump #7 was rebuilt. On Screw Pump #7, the upper and lower motor bearings were replaced and the defective screw pump gear box was rebuilt using all new bearings, shafts, sleeves and seals. In February, Raw Sewage #3 Pump was rebuilt and impeller balanced. The failed motor drive of ABF Pump #1 was rebuilt and upper and lower motor bearings were replaced. The defective center drive of Secondary Clarifier #3 North was replaced with a new drive and the sweeper arms were replaced and leveled. In March, at the lagoon, the south submersible pump was rebuilt and new impeller, bearings and seals were installed. The road around Lagoon #2C and #3 Lagoon was repaired with gravel from the stock pile. A contractor repaired the broken 18 inch primary sludge line, including 3 holes with 18" x 30" JCM clamp coupling, and replaced 4 sections of 8" x 20' pipe.

In April, a new stuffing box assembly was fabricated and installed on Raw Sewage pump #4. The Raw Sewage Pump #1 Pump shaft sheared off, bending mag drive shaft. The impeller was balanced, pump rebuilt and mag drive repaired and reassembled. #1 Lagoon west dog house pump station was rebuilt. The motor frame was re-welded and new drive belt installed on pump. In May, the Blower #1 south switch gear was installed and blower put back into use after being out of service for 2 ½ years. The Lagoon Submersible Tanks #1 and #2 were dismantled and old worn concrete was recoated. Screw Pump #1 bottom end cap broke away from flight assembly. Sleeve inserts were fabricated and plating was welded on gussets. The 20,000 pound flight assembly was removed and broken lower end cap repaired. The entire assembly was refurbished, sandblasted, primed and two-part epoxy coating applied. The upper rotating unit was rebuilt and lower rotating unit fabricated. A flight assembly was installed putting the screw pump unit back into service. At Lagoon #2A Pump Building, all 6 sludge pumps were rebuilt, rotors and stators changed out, and head units, ring and ball bearings replaced. In June, the entire dump bed of E3014 dump truck was replaced and dump bed cylinder rehabbed. At the Maintenance Shop, old outdated garage lift was removed and replaced with new style Mohawk lift. The floor and walls were epoxy coated. The crane track and iron work was primed and painted. A new gasoline fuel tank and used oil tank including a new fuel tank pad large enough to accommodate fuel tank catch-pit and tank, new used oil catch-pit and tank, were all installed. A canopy was fabricated, installed, and equipment placed in service. A new stuffing box assembly for Raw Sewage Pump #1 was fabricated and installed. Due to all bolts shearing off at top where rotating unit bolts to flight assembly, 20 broken bolts were extracted from Screw Pump #4 and upper rotating unit was rebuilt. Return Sludge Motor #2 North was rebuilt due to failed motor drive and upper and lower motor bearings were replaced.

In July, Primary Clarifier # 1 weir trough was sand blasted and recoated and odor control covers were rebuilt. A gate for road going to Manhole #9 was built and a road built. Hydraulic system plungers on dewatering belt filter presses #6 and 8 were replaced. The upper rotating unit of Screw Pump #5 was completely rebuilt, including shaft repair, new bearings, oil seals spacers and locking rings. Eight (8) sheets of 3' x 20' Panel roofing were replaced on Secondary Dewatering Building. Raw Sewage #1 motor, mag drive, and pump were rebuilt and impeller balanced. ABF Motor #6 motor was rebuilt due to a failed motor drive and upper and lower motor bearings were replaced. Fine Bar Screen #2 was removed and entire rake assembly was rebuilt and reinstalled. In August, new mixing valves were installed on Dewatering Belt Presses #3, #4, and #5. ABF Motor #4 was rebuilt due to failed motor drive and upper and lower motor bearings were replaced. Approximately 750 feet of perimeter fence at the south east side of admin building was replaced. All return rollers were replaced on all belt filter presses. The drain pit southwest of #5 Lagoon was cleaned out using badger to daylight and vactor the pit. Rusted old carbon steel sweep arms were removed from #1 Primary Clarifier and newly fabricated stainless steel arms and scum flaps were installed. Secondary Clarifier #3 South broken center column was replaced, new center drive and stainless steel sweep arms were installed, the center drum was sanded, primed and painted, and newly refurbished cat walk installed. The Biogas Control Room HVAC was beyond repair and replaced with a new 4-ton split system HVAC. In September, the road to the Horn Lake Metering Station was repaired by installing 12' x 20' cross over drain line. #1 Primary Clarifier was sandblasted and epoxy coating applied, sweep arms were replaced with newly fabricated stainless steel arms, and flaps and sweep arm paddle replaced. #6 ABF Motor was rebuilt due to failed motor drive and upper and lower motor bearings replaced. #3 Final Effluent Pump Motor windings were cleaned, recoated, and new bearings installed. #1 Raw Sewage Pump was completely rebuilt with new pump shaft, bearing, seal and wear rings and new stainless steel packing gland insert was fabricated and installed on pump housing. #2 Screw Pump was completely rebuilt, including pump shaft, bearing and seal. Lagoon Maintenance Shop Building was pressure washed, primed and repainted.

In October, #2 Final Effluent Pump Motor windings were cleaned, recoated, and new bearings installed. Screw Pump #2 bottom end cap broke away from flight assembly. New sleeve inserts and plating were fabricated and welded on gussets. The 20,000 pound flight assembly was removed and broken lower end cap repaired. The entire assembly was refurbished, including sandblasting and priming and a two-part epoxy coating was applied. The upper rotating unit was rebuilt and flight assembly installed. #5 Cell in in the north 110 Acres was built up and cut to drain to the south. #4 Final Effluent pump Motor windings cleaned, recoated, and new bearings installed. Audio visual equipment was installed in the Admin Building conference room. In November, the Lagoon primary road was graveled using CR610 from the stock pile. Lagoon #6 Cell was cleaned out. #1 Screw Pump was rebuilt, including upper rotating unit, shaft repair, new bearings, oil seals spacers and locking rings. In December, #1 South Blower Motor was cleaned, babbitt bearing repaired and oil seals were replaced. The north side of #5 Lagoon was cleaned out with assistance from City Heavy Equipment. Trees were removed to begin construction of access for #9 man hole.

In addition to the major tasks listed above, the Maintenance Department completed 3417 Preventative Maintenance Work Orders, 799 Scheduled Maintenance Work Orders, and 1190 Unscheduled Maintenance Work Orders for a total of 5416 work orders (including some not categorized). More work orders were entered into MP2 compared to previous years due to faster internet connection speeds. The annual number of work orders completed over the last 10 years is summarized as follows:

| Year | 2018 | 2017 | 2016 | 2015 | 2014 | 2013 | 2012 | 2011 | 2010 | 2009 |
|---|---|---|---|---|---|---|---|---|---|---|
| Preventative | 3417 | 1896 | 178 | 567 | 1000 | 1236 | 1777 | 1285 | 1661 | 3226 |
| Scheduled | 799 | 321 | 20 | 72 | 96 | 165 | 263 | 185 | 109 | 188 |
| Unscheduled | 1190 | 252 | 18 | 118 | 125 | 138 | 596 | 658 | 739 | 600 |
| Total | 5416 | 2469 | 265 | 772 | 1234 | 1784 | 2650 | 2128 | 2509 | 4014 |

Ferrous Chloride Use –No ferrous chloride has been used at the Maxson Facility since January 2012. Ferrous chloride reduces the likelihood of struvite formation by producing acidic conditions and binding up magnesium and phosphorus ions. Struvite formation began to be observed in Lagoon Nos. 3 and 4 in 2015, favoring areas where there is turbulence. A 5000 gallon tank has been placed into position to add ferrous chloride to feed sludge prior to dewatering when this practice is resumed in this more limited manner. No ferrous chloride was purchased or used in 2018.

Flow – The average daily flow rate in 2018 was 78 mgd which was 10 mgd higher compared to the previous year. The 0.5$^{th}$ percentile flow in 2018 was 62 mgd which is the expected domestic flow. The average daily flow received from significant industrial users was 9.6 mgd. This leaves 6.4 mgd which is 8.2% of average daily flow unaccounted for. The total rainfall for the year was 64 inches (NWS). Memphis annual rainfall has averaged 52 inches.

$BOD_5$ – The average influent $BOD_5$ loading for 2018 was 535,000 pounds per day, which is 84,253 pounds or nearly 19% increase from the previous year. The average influent $BOD_5$ concentration was 829 mg/l and the average effluent concentration was 40 mg/l, resulting in a removal efficiency of 95%. The total average $BOD_5$ reported being discharged by the nine largest industrial users was 265,305 pounds per day in 2018, equivalent to 50% of the total loading. This is 18,730 pounds or 7.6% more in monitored industrial loading from the previous year. The 0.5$^{th}$ percentile loading received in 2018 was 175,311 ppd which is expected domestic loading. This leaves 94,384 ppd of influent $BOD_5$ which is 18% of average influent BOD loading unaccounted for.

Total Suspended Solids (TSS) – The influent TSS averaged 526,650 pounds per day, which is 99,335 pounds or a nearly 23% increase from the previous year. The average influent TSS concentration was 807 mg/l and the average effluent concentration was 47 mg/l resulting in a removal efficiency of 94%. The total average TSS reported discharged by the nine largest industrial users was 171,316 pounds per day in 2018, equivalent to 32.5% of the total loading. This is 20,117 pounds or 13% increase in monitored industrial loading from the previous year. The 0.5$^{th}$ percentile loading for 2018 was 140,135 ppd which is expected domestic loading. Lagoon supernatant and dewatering filtrate return was calculated to be 51,692 ppd. This represented nearly 10% of influent TSS received in 2018. This leaves 163,507 ppd of influent TSS which is 34% of influent TSS loading unaccounted for.

Chemical Oxygen Demand (COD) - The influent averaged 1602 mg/l and 1,066,357 ppd. The effluent averaged 227 mg/l and 147,668 ppd. The average removal was 86%. The 0.5$^{th}$ percentile loading for 2018 was 139,046 ppd which is expected domestic loading. The total average COD discharged by the 9 largest industrial users was 566,481 ppd which equals 55% of the total loading. The remaining 360,830 ppd or 34% of influent COD is unaccounted for.

Effluent E. coli - E. coli analysis was performed on daily effluent grab samples throughout the year as required by the NPDES permit as report only. Effluent E. coli results for this past year are presented in Figure 1 with apparent seasonal trends.

Water – In 2018 metered water usage was 58,915 CCF per month resulting in total usage of 589,153 CCF. The total annual cost was $544,078. Secondary Dewatering potable water meter malfunctioned in April 2018 so no data was available for the rest of the year. The estimated water usage for 2018 was 83,158 CCF per month resulting in a total estimated usage of 997,896 CCF or 2 MGD. This is an increase of 3997 CCF per month from the previous year. The estimated water 2018 water cost was $1,010,064 if it all had been billed.

Energy – The main power consumption of the plant was an average of 5,652,980 kwh per month in 2018. This is 53,327 kwh more per month than the previous year. The monthly peak demand averaged 8728 kW which is 52 kW more than the previous year. The average monthly main power bill was $400,017, which is $2463 less than the previous year. Payments for participating in the ENERNOC curtailment program for main power totaled $58,802.60. September was the high month for demand at 10,329 kw and for consumption at 5,980,300 kwh. June was the low month for demand at 8042 kw and February for consumption at 5,255,950 kwh. Hourly power consumption throughout 2018 is shown in Figure 2.

Effective 10-1-2015, the City entered into an agreement with MLGW to switch from Seasonal Demand to Time of Use billing. TVA would charge Fuel Cost Adjustment based on user class. TOU rate structure benefits high load factor customers through incentivizing off-peak energy usage. The contract term is 5 years and will be automatically extended unless otherwise terminated.

Sludge and Biosolids – The quantity of primary sludge wasted to the lagoons in 2018 was 113.5 million, an increase of 12.5 million pounds from the previous year. The quantity of secondary sludge wasted to the lagoons in 2018 was 24.8 million pounds, an increase of 2.6 million pounds from the previous year. The total quantity of both primary and secondary sludge wasted from the plant was 138.3 million pounds, an increase of 15.1 million pounds from the previous year.

Approximately 23 million pounds of digested sludge was dewatered at the secondary dewatering facility and surface disposed as bio-solids at a designated area within the plant site. This includes an unknown amount of ferrous chloride residual from previous years that was added to Lagoon Nos. 3, 4, and 5 (the uncovered lagoons). Approximately 49.2 % of the combined primary and secondary sludge wasted to the lagoons in 2018 was destroyed equivalent to roughly 68 million pounds. The formula used to calculate percent destroyed is %VS of wasted primary and activated sludge multiplied by %VAR. The %VS for primary and waste activated sludge is based on two separate sampling points and then combined mathematically based on proportion. The %VS for feed sludge is based on a single sampling point.

A total of 91 million pounds of digested sludge was disposed in 2018 as surface disposed biosolids (11,500 tons) and volatilized lagoon sludge (34,000 tons) at a total cost of approximately $3,049,374. This equates to a unit cost of $67 per dry ton. The annual cost per dry ton for surface disposal and volatilization combined for the past 10 years is as follows:

| 2018 | 2017 | 2016 | 2015 | 2014 | 2013 | 2012 | 2011 | 2010 | 2009 |
|---|---|---|---|---|---|---|---|---|---|
| $67 | $66 | $76 | $68 | $68 | $62 | $68 | $59 | $49 | $53 |

Generally, the unit cost to dewater and dispose sludge as bio-solids is a function of polymer cost and dewatering throughput. The cost to dewater digested sludge and surface dispose as bio-solids (i.e. dewatered digested sludge) in 2018 is estimated to be $265 per dry ton. The annual cost to dewater digested sludge and surface dispose as bio-solids per dry ton for the past 10 years is as follows:

| 2018 | 2017 | 2016 | 2015 | 2014 | 2013 | 2012 | 2011 | 2010 | 2009 |
|---|---|---|---|---|---|---|---|---|---|
| $265 | $222 | $201 | $214 | $262 | $190 | $247 | $145 | $100 | $131 |

The annual amount of digested sludge (in million dry pounds) removed from the lagoons over the last 10 years is provided below:

| 2018 | 2017 | 2016 | 2015 | 2014 | 2013 | 2012 | 2011 | 2010 | 2009 |
|---|---|---|---|---|---|---|---|---|---|
| 23 | 24.7 | 29.95 | 27.2 | 23.4 | 31.2 | 24.1 | 41.67 | 56.3 | 44.9 |

For the last 10 years, net accumulation has been occurring at an average rate of 42.3 million dry pounds per year. Surface disposal has averaged 36.2 million dry pounds per year (assuming 90% capture). In order to stop net accumulation, the Maxson Facility will need to dewater and surface dispose an average of 78.5 million dry pounds per year which is more than double the existing rate.

Biogas – For the year 2018, the Maxson Facility produced a total of 565 million cubic feet of biogas. TVA used zero million cubic feet of the annual production.

In August 2014, the TVA board of directors decided to construct a 1,070-megawatt combined-cycle gas-fired plant that will replace the three existing coal-fired units at the Allen Fossil Facility located adjacent the Memphis Facility by December 2018 to comply with a 2011 agreement with the U.S. Environmental Protection Agency. The combined-cycle gas-fired plant went into service on 4-30-2018 at a total cost of $975 million.

The annual biogas produced by the Maxson Facility for the past 10 years is as follows (in millions of cubic feet):

| 2018 | 2017 | 2016 | 2015 | 2014 | 2013 | 2012 | 2011 | 2010 | 2009 |
|---|---|---|---|---|---|---|---|---|---|
| 565 | 744 | 1167 | 1254 | 1155 | 1027 | 1085 | 1015 | 851 | 1160 |

The annual daily average biogas produced for the past 10 years is as follows (in millions of cubic feet):

| 2018 | 2017 | 2016 | 2015 | 2014 | 2013 | 2012 | 2011 | 2010 | 2009 |
|---|---|---|---|---|---|---|---|---|---|
| 1.55 | 2.04 | 3.20 | 3.44 | 3.16 | 2.81 | 2.97 | 2.78 | 2.33 | 3.18 |

The biogas produced equates to 8.3 cubic feet of biogas per pound of volatiles destroyed (12 to 18 cubic feet of biogas per pound of volatiles destroyed is the standard range for digester vessels.) Annual biogas yield for the last 10 years (ratio of biogas to volatiles destroyed) in cubic feet per pound is as follows:

| 2018 | 2017 | 2016 | 2015 | 2014 | 2013 | 2012 | 2011 | 2010 | 2009 |
|---|---|---|---|---|---|---|---|---|---|
| 8.3 | 12.4 | 23.8 | 21.4 | 14.1 | 12.5 | 13.2 | 11.2 | 14.7 | 17.4 |

To a certain point, the biogas yield is directly related to the ratio of primary to waste activated sludge as shown in Figure 3. It is believed that too high of a primary to waste activated sludge ratio was a factor in the drop in biogas yield and production since 2017 as well as the build-up of inert solids and short circuiting in Lagoon 2A. Increasing biogas production will involve both reducing the ratio of primary to waste activated sludge and restoring normal conditions in Lagoon 2A.

The Maxson Facility flare is currently permitted as a "conditional minor" source. A Construction Permit for the new flares was submitted to the SCHD on 11-23-2016. A revised permit application was submitted on 1-17-2017 to include the biogas flows and emissions to conform to CDM Smith's design numbers. The SCHD issued a construction permit on 4-10-2017, expiring on 10-11-2018. The City provided comments to the SCHD regarding the permit conditions on 5-5-2017. The City transmitted a Title V operating permit application to the SCHD on 5-18-2018 due to the Major Source designation based on potential air emissions of CO in excess of 100 tpy. Deficiencies were noted in the application submittal and subsequently corrected on 6-11-2018. As a result of the Title V submission, the construction permit would no longer be applicable.

The annual biogas usage by TVA along with the amount of biogas flared as waste gas by the Maxson Facility over the last 10 years is as follows (in millions of cubic feet):

| Year | 2018 | 2017 | 2016 | 2015 | 2014 | 2013 | 2012 | 2011 | 2010 | 2009 |
|---|---|---|---|---|---|---|---|---|---|---|
| TVA | 0 | 0 | 0 | 0 | 60 | 131 | 397 | 414 | 106 | 347 |
| Flared | 565 | 744 | 1167 | 1254 | 1094 | 896 | 688 | 574 | 746 | 813 |

The City received zero in revenue from TVA for the Maxson biogas for 2018.

Permit Excursions – This year the Maxson Facility had 53 excursions of its NPDES permit limits related to treatment efficiency and performance. The excursions are described as follows:

In February there were 5 permit excursions. There was one monthly average effluent BOD loading excursion. There were one daily concentration, one monthly average concentration, one monthly average loading, and one weekly average loading TSS excursions. This was attributed to a failed final clarifier drive and a failed final clarifier drain lift station.

In March, there were 16 permit excursions. There were five (5) daily concentrations, one weekly average concentration, one monthly average concentration, one weekly average loading, and one monthly average loading BOD excursions. There were two (2) daily concentration, one weekly average concentration, one monthly average concentration, two (2) weekly average loading, and one monthly average loading TSS excursions. This was attributed to elevated flows due to high Mississippi River stages.

In August, there were 24 permit excursions. There were four (4) daily concentration, one weekly average concentration, one monthly average concentration, one weekly average loading, and one monthly average loading BOD excursions. There were four (4) daily concentration, one weekly average concentration, one monthly average concentration, one weekly average loading, and one monthly average loading TSS excursions. There were three minimum effluent settleable solids and three effluent minimum dissolved oxygen excursions. There were two (2) daily minimum removal TSS excursions. This was attributed to a broken primary sludge wasting line.

In December, there were 8 permit excursions. There were three (3) daily concentration, one weekly average concentration, one average monthly concentration, one weekly average loading, and one monthly average loading TSS excursions. There was one minimum daily effluent dissolved oxygen level excursion. This was attributed to blower malfunction and no working backup.

The annual number of permit excursions related to treatment efficiency and performance over the years is provided in Figure 4. During years of stable operation (approximately 22 of 36 years), the Maxon Facility has averaged 3.3 violations per year. During remaining years, violations have mainly occurred due to construction projects, heavy rains or collection system failures at an average of 34.2 violations per year. The Maxson Facility has had two years (1987 and 2010) with zero violations.

Flood Response Activities

During the April 28 through June 3, 2011 period, Maxson Facility personnel performed additional tasks unrelated to normal plant operations. The Maxson Facility's direct costs related to flood response consisted of $121,046.25 for 3031 overtime hours and $156,837 for expenditures. Indirect costs consisted of $277,883 in equipment usage (i.e. mobile equipment such as portable pumps, dozers, etc.). Worksheets documenting these costs were submitted for reimbursement under FEMA Event 1979. Secondary Clarifier 3N Stabilization project at the Maxson Plant was completed on 12-9-2013 for $548,500 and submitted as an additional FEMA project. Total cost for the additional FEMA project was $599,631.53, which includes outside engineering, materials, and force account. Total approximate costs for flood response activities eligible for submittal to FEMA for reimbursement were $1,155,397.80. On 4-11-2013, $231,843 reimbursement was received from FEMA toward these costs. On 2-8-2017, reimbursement of $599,402 was received. On 5-8-2017, final payment of $34,040 was received for a total reimbursement of $865,285 from FEMA or 75% of total costs. The accounting may be slightly off due to reimbursement overlap between services centers.

Disinfection, Process Improvements, and Odor Control

In 2014, the disinfection, process capacity and treatment improvements, and odor control projects were combined into a single initiative called Construction Manager at Risk (CMAR). Treatment capacity increase was based on an average daily flow increase from 70 to 90 mgd at influent BOD concentration of 800 mg/l. The equivalent rated BOD capacity should be in the vicinity of 648,000 ppd once capacity increases and performance improvements have been made.

During the year, Package 1 covering disinfection improvements advanced from 90% design to contract execution and Guaranteed Maximum Price #1 in the amount of $43,361,322 issued for construction. $25 million of the package was covered by an SRF loan secured later in the year. Notice to Proceed was issued on 7-31-2017 with a deadline of 7-31-2019 for completion. Pre-construction meeting for Package 1 was held on 3-12-2018 after months of delay of release of funding by SRF. The final Package 1 was approved by SRF on 5-25-2018. The coarse bar screen replacement designated as Package 2 was approved by SRF on 3-19-2018 and Guaranteed Maximum Price #2 in the amount of $10,722,147. A NTP was issued to CDM Smith on 6-4-2018. The final Package 2 was approved by SRF on 9-12-2018.

During the year, Package 2A covering secondary clarifiers, intermediate pumping improvements, and fine bubble diffusers advanced to 100% design and approved by SRF. Package 2B covering re-aeration tanks and a new blower building advanced to 100% design and then valued engineered. Package 3 which covered bio-tower media and odor dispersion fans advanced to 60% design and then was broken into separate Packages 3A and 3B to facilitate procurement.

Biogas Quality Investigation

Two (2) samples of biogas were collected throughout the year and analyzed for hydrogen sulfide and BTU. Hydrogen Sulfide concentrations averaged 4,540 ppmv ranging from 680 to 8,300. Energy values averaged 350 BTU/cubic feet (wet) ranging from 310 to 390. Average annual values of analytical results from previous years are summarized as follows:

| Year | Soloxanes (ppbv) | Hydrogen Sulfide (ppmv) | BTUs/cubic foot (wet) |
|---|---|---|---|
| 2012 | 250 | 8451 | 440 |
| 2013 | 124 | 19,820 | 468 |
| 2014 | 156 | 4591 | 441 |
| 2015 | 152 | 2836 | 483 |
| 2016 | NA | 935 | 320 |
| 2017 | NA | 1981 | 420 |
| 2018 | NA | 4540 | 350 |

TVA Conversion to Natural Gas Related Projects

In August 2017, installation of the 5 new flares was completed by Kiewit Construction under TVA's contract and flaring of the waste gas using the new flares initiated. In September 2017, installation of the 4 bioreactors was completed and the reactors charged with media. Kiewit's contract to complete the job ended 11-5-2017 and operation turned over to the City. The City continued to work with CDM Smith and Biorem throughout the year to start-up the bio-reactor and achieve steady-state operation.

In January, it was noted that the heat exchange plates were getting clogged and had to be cleaned. In February, it was observed that recirculation water was 70 degrees F and not at the required 95 degrees F. In March, it was noted that potable water supply pressure was possibly too high. In April, Kiewit was released from the project by TVA and CDM Smith replaced 2-

inch with 3-inch water piping. In May, CDM Smith completed installation of the larger piping, replaced sensors, and added PLC to keep from overfilling the bio-reactors. In June, a warranty claim was filed because the heat exchanger could not maintain the required 95 degrees F and the heat exchanger subsequently replaced. The City put Biorem on notice that the system was non-performing with a 7-9-2018 deadline to comply and Biorem subsequently re-seeded Bioreactor #2 with new microbes. It was observed that recycle pumps for Bioreactors #2, #3, and #4 were not working. CDM Smith ordered a hydrogen sulfide meter that could read up to 10,000 ppm which apparently the bioreactor was treating. In August, Biorem retrofitted Bioreactor #2 to allow removal of heat exchanger while in service and #2 recycle pump failed. In September, Maxson Facility personnel had to bypass the coalescing filters so that biogas could go directly to the flares while the bioreactors were being worked on. Three reactors were returned to service in October with at least one meeting hydrogen sulfide specifications. Warranty claims were pursued against TVA and Kiewit. In November, two of the reactors were in service and occasionally meeting required specifications however recirculation pumps were tripping out due to clogging in the heat exchangers. In December, three of the reactors were in service and occasionally meeting specifications. The City continued efforts to transition from Dragger Tubes and hand held units to on-line analyzer for hydrogen sulfide monitoring.

The revenue contract was signed with TVA on 1-3-2018 with a 20 year term effective once biogas treatment is adequate. The estimated capital contribution is $20 million based on a 6,500,000 MMBtu minimum agreement volume with TVA concurrently agreeing to pay the City 33 cents per MMBtu spread out over a 13-year payback period. This is based on a minimum annual average 500 BTU wet (higher heating value) and maximum monthly average Hydrogen Sulfide concentration of 30 grains per 100 CF or 480 ppm.

The Alan Fossil Facility was retired on 3-31-2018. TVA plans is to demolish the Allen Fossil Facility 6 years after operation of the new combined cycle plant begins.

## 2.0 Stiles Wastewater Treatment Facility

Introduction-

We sadly report the passing of the Stiles Facility namesake Maynard C. Stiles on 11-12-2017 at age 89. He worked for the City of Memphis from 1967 to 1989. He testified before congressional subcommittee in the late 1970s regarding the need for higher permit limits for treatment plants that have high industrial loading. At the time, the Stiles Facility was the most expensive project undertaken by the City. The Stiles Facility has served the City without process change or major modifications for 40+ years which is a rarity for any treatment plant.

A preconstruction meeting for the disinfection facility and related improvements was on 1-4-2018. American Yeast initiated a project to construct terminal and pipe line to receive molasses for unloading by barge which should result in significant cost savings (1 barge = 69 tanker trucks.) Radioactive source material at Secondary Dewatering was removed on 2-6-2018 by a licensed disposal contractor getting the Stiles Facility out of the nuclear storage business. Kick-off meeting for new Stiles Lab was on 2-8-2018. A new parking lot with 108 parking spaces was constructed during April and May as part of the disinfection facility project. During June and July, cracks were repaired in the disinfection tank dividing wall in order to keep each half dry when the other taken out of service. Vertical baffles were installed on the disinfection tank by the contractor during 6-25-2018 thru 9-24-2018 for the purpose of promoting longitudinal mixing of PAA as determined by CFD modelling. New perimeter fencing was installed during July and August as part of the disinfection facility.

PXC began construction of the Wolf River Plant on 1-10-2018 with construction of a new entrance gate closer to the Stiles Facility Admin Building. By mid-February, construction of their main plant was halted due to no formal approval from FAA. PXC continued installation of the disinfection facility (at the west end of the Stiles Facility) in order to meet their contract deadline to begin supplying and the Stiles Facility to begin using PAA for disinfection later in the year.

The City met with airport officials on 2-27-2018 and found out the restrictions were based on a planned future runway extension by 500 feet and not current conditions, a "plan" that had been in effect for the past 15 years. On 3-1-2018, airport officials offered suggestions regarding lowering tank height and shifting location to be in compliance with air space restrictions. On 8-2-2018, the FAA issued for 30 day public notice a request by PXC to shift the Wolf River Plant 90 feet further west which was eventually approved. On 8-6-2018, the City met with airport officials to consider moving to a new location which would allow the area to be used as a sewer equalization basin, potential other uses related to wastewater treatment and eliminate FAA imposed restrictions on Stiles Facility operations. It appears that the airport will remain the same size until being relocated or closed within the next 10-15 years given the current/potential expansion needs of the City.

A raw sewage pumping failure occurred on 3-9-2018 during high Mississippi River stages that would rank in the top catastrophic episodes of the Stiles Facility. The failure resulted in the overflow of sewage from manholes adjoining the Dewitt Spain Airport. A contractor (Xylem) was mobilized to pump the sewage over the levee so it would not submerge the airport while City crews pumped the flooded dry well to install a backup motor (within 58 hours) and recover the submerged motors for rewinding and re-installation. Full pumping was restored within 197 hours of the emergency. During 3-9 to 3-18-2018, 665 million gallons of sewage was pumped over the north levee and 38 million gallons pumped out of the dry well. During 3-19 to 3-30-2018, 64 million gallons was pumped back into the collection system, including 600,000 gallons meant for discharge into the collection system that was released by the airport in the Wolf River by mistake. Offending surface residue comprising approximately 25 acres adjoining the airport runway to the west and south was subsequently scraped and relocated to Lagoon 3 North in consultation with TDEC. The area disturbed by the emergency pumping north of the runway was restored by Heavy Equipment to the satisfaction of the airport by 8-8-2018. The City received $464,299.83 from FM Global for costs associated with restoring raw sewage pumping capacity. No costs related to protecting the airport from flooding were eligible for reimbursement. A table of previous raw sewage pumping emergencies is provided as Figure 5.

In April, odor complaints started coming in mainly due to wasting of fresh activated sludge into uncovered lagoon 3A and possibly due to residue from the sewer overflow during the sewage pumping emergency. The odor complaints appeared to subside by June after stabilization of the wasted sludge had occurred within the open lagoon. Approximately 10 loads of agricultural lime were added just to be sure. SARP10 removed three sewer plugs from the headworks during June and July which had been lost during a 5-inch rainfall on 8-31-2017.

A kick-off meeting was held on 8-17-2018 with Barge Design Solutions regarding Pump Station Improvements and other miscellaneous tasks including review of a hydraulic model of Stiles Facility sewer flow, nutrient recovery at the Maxson Facility, upcoming sampling of Lagoon Nos. 1 and 2, and the development of Preliminary Engineering Report implementing recommendations of previous biosolids master planning workshops.

A PAA Disinfection Facility commissioning review was performed on 8-23-2018 as the facility approached completion. The first load of PAA was delivered on 9-11-2018 and initial dosing began on 9-12-2018. The dosing optimization period was shorter than expected as PXC relied on their previously developed algorithm with little additional changes. Round the clock dosing with PAA began on 11-10-2018 after contractor completed installation of vertical baffles in the disinfection tank, relying on continuous truck deliveries to supply the need since the Wolf River Plant was not in operation as originally planned. The vertical baffles, installed for the purpose of reducing PAA demand in the disinfection tank, also proved effective at trapping floating scum during November and December as temperatures dropped.

The Maintenance Department performed numerous noteworthy tasks during the year. At the Influent Pump Building, new water resistant doors were installed in the Joy Fan Room, Header Room, and influent wet well entries. KSB Pump was installed in #1 hole in Pump Building. Back up pumps were staged at the Influent Pump Building for flood season. The #1 Joy Fan was rebuilt and put back in service. The bad HPU solenoids on #4 KSB check valve were changed out. A rebuilt Bently Nevada Vibration monitor rack for the KSBs was installed to replace a failed older unit.

At the Grit Tanks, new chutes were installed for North and South Vulcan wash presses. New liners were installed in North and South Screw Conveyors. #4 Drag Out was completely rebuilt, the bed plate was upgraded from 3/8" to ¾" plate, and a new weir wall was installed. A new drive sprocket and chain were installed on #1 Drag Out.

At the Aeration Basins, a leak on east side of Aeriation Tank was injected and repaired. New spray water pump bases were fabricated and installed. New diffusers in B-1 and A-6 Aeriation Basin were installed. Approximately 10 new diffusers were added in Final Channels to replace bad Swing Fusers. D.O. probes were replaced with new units.

At Return Sludge, a new WEMCO Pump was installed on B side. At the Blower Building, a new Joy Fan Motor was installed at #3 Blower Motor. New annunciator panels were installed on #2 and #4 Blower Control cabinets for alarms. Four (4) limitorque actuators in the Air Main were reconditioned. At the Final Tanks, new level controllers were installed on the Scum Wells.

At Secondary Dewatering, new steel panels were installed on back of loadout bay. New conduit and wiring was installed for new air compressors. The old A/C unit was disconnected and new A/C unit installed. Temporary electrical feeds were installed for pilot testing new dewatering equipment.

Thermal Imaging was performed on all Switchgear. Old fluorescent lighting was changed out and replaced with LED lighting throughout the Admin Building.

In addition to these major tasks, the Maintenance Department completed 2795 Preventative Maintenance Work Orders, 758 Scheduled Maintenance Work Orders, and 255 Unscheduled Maintenance Work Orders for a combined total of 3863 Work Orders (including some not categorized). The annual number of work orders completed over the last 10 years is summarized as follows:

| Year | 2018 | 2017 | 2016 | 2015 | 2014 | 2013 | 2012 | 2011 | 2010 | 2009 |
|---|---|---|---|---|---|---|---|---|---|---|
| Preventative | 2795 | 3148 | 2694 | 2738 | 2802 | 2832 | 2402 | 2563 | 2741 | 2805 |
| Scheduled | 758 | 497 | 345 | 552 | 799 | 837 | 1049 | 781 | 941 | 925 |
| Unscheduled | 255 | 188 | 138 | 156 | 192 | 323 | 364 | 325 | 432 | 335 |
| Total | 3863 | 3857 | 3177 | 3446 | 3810 | 3992 | 3747 | 3669 | 4114 | 4065 |

Flow – The average daily influent flow for the Stiles Facility in 2018 was 79 MGD. This is an increase of 4 MGD from the previous year. The 0.5$^{th}$ percentile flow in 2018 was 47 mgd. The average daily flow from the major significant industrial users was 14.9 mgd. This leaves 17.1 mgd which is 21.6% of average daily flow unaccounted for. The rainfall was 62.5 inches (plant gage). The average annual rainfall at the Stiles Plant gage has been 55 inches.

$BOD_5$ – The average influent $BOD_5$ in 2018 was 268,114 pounds per day. This is 20,763 pounds or 8% more loading from the previous year. The average influent $BOD_5$ concentration was 417 mg/l and the average effluent concentration was 32 mg/l for a $BOD_5$ removal efficiency of 92%. The total average $BOD_5$ in pounds per day reported discharged by the seven largest industrial users was 169,779 pounds per day in 2018 which equals to 63% of the total loading. This is 7876 pounds or 5% more than the previous year. The 0.5$^{th}$ percentile loading in 2018 was 94,476 ppd which is estimated domestic loading. This leaves 3859 ppd or 1.4% of influent BOD loading unaccounted for.

Total Suspended Solids (TSS) – The influent total suspended solids averaged 294,417 pounds per day in 2019. This is 10,523 pounds or 3.6% less than the previous year. The average influent TSS concentration was 456 mg/l and the average effluent concentration was 29 mg/l for a TSS removal efficiency of 94%. The total average TSS reported discharged by the nine largest industrial users was 64,697 pounds per day in 2018 which equals to 22% of the total loading. This is 1645 pounds or 2.5% more from the previous year. TSS loading from lagoon supernatant and dewatering filtrate return was calculated to be 43,032 ppd or 14.6% of total average daily TSS loading. The 0.5$^{th}$ percentile loading for 2018 was 105,473 ppd which is estimated domestic loading. This leaves 81,215 ppd of TSS or 27.6% of average daily loading unaccounted for.

Chemical Oxygen Demand (COD) - The influent averaged 958 mg/l and 607,933 ppd. The effluent averaged 200 mg/l and 129,403 ppd. The average removal was 79%. The 0.5$^{th}$ percentile loading for 2018 was 147,749 ppd which is estimated domestic loading. The total average COD discharged by the 8 largest industrial users was 409,255 ppd which equals 67% of the total loading. This leaves the remaining 50,929 ppd or just over 8% of influent COD loading unaccounted for.

Effluent E. Coli - Effluent E. coli analysis was performed on daily effluent grab samples throughout the year as required by the NPDES permit as report only. Effluent E. coli results for this past year are presented in Figure 6 with impact of effluent disinfection shown toward the end of the year.

Water – In 2018 the metered water usage totaled 353,480 CCF at a cost of $352,730. The monthly average metered water consumption was 35,348 CCF. These amounts and the amounts reported in previous years are low due to a problem with potable water metering that was corrected in March 2018. Actual annual potable water consumption is estimated at 501,600 CCF or 1 mgd. A project is being planned in 2019 to meter diverted well water from American Yeast so that a complete accounting of all water use can be made that will serve as a baseline for future water conservation efforts.

Energy – The plant main power consumption was an average of 5,468,778 kwh of electricity per month in 2018. This is approximately 405,163 kwhs per month more than the amount used in the previous year. The monthly peak demand averaged 8475 kw which is 604 kw more than the previous year. The average monthly main power bill in 2018 was $391,714. This is $30,430 more per month than the previous year. Payments for participating in the ENERNOC curtailment program for main power totaled $32,376.74.

January was the high month for demand at 8814 kw. December was the high month for consumption at 5,972,104 kwh. May was the low month for demand at 8168 kw and March was the low month for consumption at 4,844,898 kwh. Daily power consumption throughout 2018 is shown in Figure 7.

Effective 10-1-2015, the City entered into an agreement with MLGW to switch from Seasonal Demand to Time of Use billing. TVA would charge Fuel Cost Adjustment based on user class. TOU rate structure benefits high load factor customers through incentivizing off-peak energy usage. The contract term is 5 years and will be automatically extended unless otherwise terminated.

Sludge and Biosolids – The Stiles Plant wasted 115.1 million dry pounds of secondary sludge to the lagoons in 2018. This is essentially no change from the previous year. 38.6 million dry pounds (based on counting trucks) was surface disposed at a 88.8% capture rate.

The annual amount of digested sludge (in million dry pounds) removed from the two covered lagoons over the last 10 years is provided below:

| 2018 | 2017 | 2016 | 2015 | 2014 | 2013 | 2012 | 2011 | 2010 | 2009 |
|---|---|---|---|---|---|---|---|---|---|
| 43.9 | 31 | 44 | 30 | 29.4 | 29 | 36 | 17.6 | 34 | 17 |

Approximately 35% of the secondary sludge wasted to the lagoons in 2018 was destroyed. The formula used to calculate percent destroyed is %VS of WAS multiplied by %VAR. This equals 39.9 million dry pounds (19,950 tons). Considering land filling and volatilization the new change in the lagoon solids inventory was an increase of 36.6 million dry pounds within Lagoon Nos. 1, 2, and 3. The calculated cumulative amount of dry solids within the three lagoons is 687 million dry pounds. In 2018, approximately 38.6 million dry pounds (19,300 dry tons) of bio-solids were disposed in the Surface Disposal Site. The calculated amount of dry bio-solids that has been surface disposed since 1984 is 1311 million dry pounds.

For the last 10 years, net accumulation has been occurring at an average rate 28.9 million dry pounds per year. Surface disposal has averaged 27.8 million dry pounds per year. In order to stop net accumulation, the Stiles Facility will need to dewater and surface dispose or land apply an average of 56.7 million dry pounds per year which is twice the existing rate.

The total cost for the dewatering and surface disposal operation was approximately $3,267,102. This equates to a unit cost of $83 per dry ton of bio-solids surfaced disposed and volatilized (combined) in 2018. The annual cost per dry ton for surface disposal and volatilization combined over the past 10 years is as follows:

| 2018 | 2017 | 2016 | 2015 | 2014 | 2013 | 2012 | 2011 | 2010 | 2009 |
|---|---|---|---|---|---|---|---|---|---|
| $83 | $93 | $85 | $102 | $70 | $78 | $67 | $57 | $86 | $79 |

Generally, the unit cost to dewater and dispose sludge as bio-solids is a function of polymer cost and dewatering throughput. The cost to dewater and surface dispose digested sludge was $169 per dry ton in 2018. The annual cost per dry ton based on dewatering throughput over the past 10 years is as follows:

| 2018 | 2017 | 2016 | 2015 | 2014 | 2013 | 2012 | 2011 | 2010 | 2009 |
|---|---|---|---|---|---|---|---|---|---|
| $169 | $250 | $168 | $129 | $199 | $192 | $162 | $189 | $129 | $232 |

Biogas- Approximately 111.7 million cubic feet of biogas was produced 2018. The annual biogas production over the last 10 years is as follows (in millions of cubic feet per year):

| 2018 | 2017 | 2016 | 2015 | 2014 | 2013 | 2012 | 2011 | 2010 | 2009 |
|---|---|---|---|---|---|---|---|---|---|
| 111.7 | 121.7 | 105.9 | 158.7 | 275 | 263.8 | 338.7 | 272.5 | 350 | 404.5 |

The annual daily average bio-gas production over the last 10 years is as follows (in million cubic feet per day):

| 2018 | 2017 | 2016 | 2015 | 2014 | 2013 | 2012 | 2011 | 2010 | 2009 |
|---|---|---|---|---|---|---|---|---|---|
| 0.31 | 0.33 | 0.29 | 0.43 | 0.75 | 0.72 | 0.93 | 0.75 | 0.96 | 1.11 |

The biogas produced equates to 2.8 cubic feet of biogas per pound of volatiles destroyed. The yield (ratio of biogas to pounds of volatiles destroyed) during the 10-year period is shown below:

| 2018 | 2017 | 2016 | 2015 | 2014 | 2013 | 2012 | 2011 | 2010 | 2009 |
|---|---|---|---|---|---|---|---|---|---|
| 2.8 | 2.8 | 2.7 | 4.7 | 8.3 | 9.9 | 10.1 | 10 | 10.7 | 16.7 |

12 to 18 cubic feet of biogas produced per pound of volatiles destroyed is the standard range for digester vessels. Reduced bio-gas production is related to reduced bio-gas yield which is the result of solids accumulation and short-circuiting in the two covered lagoons.

In 2018, American Yeast received 71.9 million cubic feet of biogas from the Stiles Facility which is nearly 64% of total biogas production. The revenue received from American Yeast for this biogas was $64,752. The annual usage by American Yeast along with the amount of biogas flared as waste gas by the Stiles Facility is as follows (in millions of cubic feet):

| Year | 2018 | 2017 | 2016 | 2015 | 2014 | 2013 | 2012 | 2011 | 2010 | 2009 |
|---|---|---|---|---|---|---|---|---|---|---|
| American Yeast | 71.9 | 85.3 | 63.1 | 80.5 | 84.3 | 99.35 | 105.7 | 76.2 | 109.6 | 62.9 |
| Flared | 39.8 | 36.4 | 42.7 | 78.2 | 191 | 164 | 233 | 196 | 240 | 342 |

Permit Excursions- This year the Stiles Facility had a 12 excursions of its NPDES Permit limits related to treatment efficiency and performance.

In January, there were 2 permit excursions. There were two (2) daily concentration BOD excursions attributed to foam and scum getting into the final effluent due to operator error.

In March, there were 9 permit excursions. There were three (3) daily concentration, one weekly average concentration, and one monthly average percent removal BOD excursions. There were two (2) daily minimum percent removal and one monthly average percent removal TSS excursions. This was attributed to high influent flow due to high Mississippi River stage. There was one by pass of treatment excursion for a period of 10 days due to raw sewage pumping failure that occurred on 3-9-2018.

In November, there was 1 permit excursion. There was one daily concentration effluent BOD concentration excursion due to cleaning scum accumulation behind the vertical baffles in the disinfection tank used for PAA mixing.

The annual number of permit excursions related to treatment efficiency and performance over the years is provided in Figure 8. During years of stable operation (approximately 20 of 36 years), the Stiles Facility has averaged 2.2 violations per year. During the remaining years, violations have mainly occurred due to heavy rains or collection system failures at an average of

47.7 violations per year. The Stiles Facility has had 5 years (1987, 1992, 1999, 2001, and 2007) with zero violations.

Root Cause Analysis of Raw Sewage Pumping Emergency on 3-9-2018

The Stiles Facility influent pumping dry well is served by two (2) Normal Duty sump pumps (both 3-inch 3-HP electric) and two backup pumps (6-inch J-line and 8-inch 50 HP Barnes, both electric). The three raw sewage pumps failed at 3:50 AM on 3-9-2018 when the three motors located within the dry well went under water. Four main breaker trips lead to this final failure. The second trip on 3-8-2018 at 4:03 PM caused the tops of manholes upstream of the plant to blow off, taking on water from the Wolf River. This increased the flow coming into the Stiles Facility. The fourth trip caused the old sampling line to fail. The root cause analysis suggested that had the old sampling line been removed much earlier and properly sealed, as intended, the failure would not have happened. Likewise, the electrical trips that led to the failure of the sampling line would not have occurred had the main breaker settings been properly made when the new switch gear was installed in 2015. The duty pumps failed and then the backup pumps failed. Had stand-by diesel pumps not dependent on electrical power been made available as a final measure when it was known that the back-up pumps were in use, the catastrophe would have been averted. The City staffed two-person crews of electricians working 12 hour shifts around the clock in order to safe guard high flow pumping (3 raw sewage pumps). At the very least, additional maintenance should have been assigned after the first trip of the main breaker on 3-4-2018 to assist with the operation of the additional equipment should problems begin to escalate.

Chapel Hill

At the Chapel Hill package treatment plant, there were no NPDES permit limit excursions. A new permit was issued 7-1-2017, effective 8-1-2017, and expiring on 7-31-2022. The City took over operation and maintenance of the Chapel Hill package treatment plant in 1972 by Council Resolution. The property was deeded from the developer to the City of Memphis on 6-28-1972, including the sewer mains and laterals. The City expends approximately $110,000 annually on operation and maintenance of that package plant. Planning was initiated in 2014 to either upgrade or replace the package plant as it is showing signs of wear. On 2-9-2015, Fisher & Arnold submitted their final report which found that the area is not in anyone's annexation reserve, the facility was basically structurally sound, the bio-towers need to be replaced and the concrete basin rehabbed. Early in 2018, the Chapel Hill influent pump and blower were connected to the Mission system by cell phone to allow remote monitoring of operations of raw sewage pumping and aeration. Discussions were initiated in 2017 to transfer operation and ownership of the package plant to Shelby County after substantial improvements have been made. The improvements are expected to be made in 2020 as a capital improvement project and if everything goes as planned, the facility will be transferred to Shelby County.

Bio-solids Long Term Planning

The current configuration of anaerobic digestion in covered lagoons, dredging and pumping to belt filter presses for dewatering to 15% solids, followed by solar drying to 50% solids in a dedicated surface disposal site a short distance away represents a very cost-effective economic model since all of this is operated and maintained in a routine manner by City personnel. The scale, reliability and effectiveness of this operation is impressive. Typically rainfall occurs at least 70 days per year in the Memphis area which impacts spreading and solar drying.

Under Pathogen Reduction for Surface Disposal, the plant has been reporting under Class B, Process to Further Reduce Pathogens, Alternative 2 (anaerobic digestion). Under Vector Attraction Reduction, the plant has reported under Alternatives 1 and 2 (greater than 38% volatile reduction, less than 17% additional volatile reduction) based on waste activated sludge as the starting point and feed solids as the end point for Alternative 1. Prior to 2002, dewatered cake was the end point for Alternative 2. Beginning in 2002, feed solids were the end point for Alternative 2.

For some historical perspective, the Stiles Facility generated 245 million dry pounds of sludge from 1978 to 1982 after the treatment plant first opened. In about 1980, the City designated a 53 acre area adjoining the south end of the airport runway as a temporary bio-solids landfill. During 1980 to 1983, the plant began dipping sludge out of the lagoons and disposing it in the 53 acre site of which only 35 acres was actually used. There was no dewatering performed at that time as the original dewatering facility was improperly designed so a second contract was issued to build a new facility. The City also issued a contract to construct Lagoon No. 4 which was completed in 1984. In the summer of 1984, the Stiles Facility was able to start sending dewatered sludge to Lagoon No. 4 which became established as the Surface Disposal Site for the next 35+ years.

Figure 9 shows the calculated cumulative net change in material within Lagoon Nos. 1, 2, and 3 and the cumulative amount of dry solids stored in the Surface Disposal Site. The Surface Disposal Site is nearing capacity and it is becoming more of a challenge to operate due to increasingly poor surface drainage.

Lagoon 3S is believed to be near sludge storage capacity based on an investigation completed during the months of November and early December 2016 with an average total solids content of 15%. Lagoon Nos. 1 and 2 are useable as long as the City can reduce the net accumulation rate. The City continued drying out Lagoon 3N using heavy equipment in order to gain use of the area as a future surface disposal site.

The City participated in the third Biosolids Master Planning workshop on 2-28-2018 followed by a kick-off meeting on 8-17-2018 that included development of a Preliminary Engineering Report for the evaluation of dewatering alternatives in conjunction of construction of a third covered lagoon to provide anaerobic digestion and Class B land application. Activities associated with the PER would include investigation of solids concentrations and quantity within Lagoon Nos. 1 and 2, bench testing of co-digestion of American Yeast beer waste with Stiles Facility, piloting screw press and centrifuge dewatering technologies, and optimizing current dewatering operations in anticipation of Class B land application.

Bench testing performed by Andritz on samples collected on 8-24-2018 using their best polymer showed that Maxson and Stiles digested sludge could achieve 17% and 22-24% cake (Maxson) and 17% and 20-22% cake (Stiles) using belt filter press and centrifuge technology, respectively. On-site visits were performed by Barge and Material Matters personnel of both dewatering facilities during 9-11 to 9-12-2018. Deficiencies noted were no flow meters for polymer or feed solids, clogged polymer injection points, and damaged or non-functioning polymer mixing valves. It was recommended that one press of each type (Andritz, Ashbrook, BDP) be outfitted with the proper flow meters and the other polymer issues be addressed before proceeding with optimization. The FKC Screw Press was pilot tested at the Maxson Facility on 9-21-2018 and at the Stiles Facility on 9-24-2018 and 9-26-2018. Cake dryness at the Maxson Facility was 24% on average based on a 0.5 to 4% feed solids range (which was variable during the pilot test). Cake dryness at the Stiles Facility was 23% on average. The largest screw press offered by FKC would have a 1250 dry pound per hour through put.

Flood Response Activities

During the April 25 through July 11, 2011 period, Stiles Facility personnel performed additional tasks unrelated to normal plant operations. The Stiles Facility's direct costs related to flood response consisted of $131,827.40 for 3528 overtime hours and $110,828.85 for expenditures. Indirect costs consisted of $242,656 in equipment usage (i.e. mobile equipment such as portable pumps, dozers, etc.). An additional claim in the amount of $145,171 was made for the replacement of the JD 160 hp track hoe. Worksheets documenting these costs were submitted for reimbursement under FEMA Event 1979. Approximate total costs for flood response activities eligible for submittal to FEMA for reimbursement were $630,483.25. On 4-11-2013, $172,493 reimbursement was received from FEMA toward these costs. On 2-8-2017, $125,696 was received. On 5-8-2017, $34,000 was received. On 10-14-2017, final payment of $50,971 was received for a total reimbursement of $383,200 from FEMA or 61% of total cost. The accounting may be slightly off due to reimbursement overlap between services centers.

Disinfection and PAA Impact on Final Effluent BOD

Sporadic full scale PAA dosing for effluent disinfection began on 9-12-2018 and continuous PAA dosing began on 11-10-2018 after the initial optimization period. The PAA usage for 2018 and calculated dosage of active ingredient is provided as follows:

| Month | Amount of 15% PAA Solution Used (pounds) | Calculated Average Dose of Active Ingredient (mg/l) | Increase in Soluble BOD Between Pre-Final and Final Effluent (mg/l) | Final Effluent BOD (mg/l) | Adjusted Final Effluent BOD (mg/l) |
|---|---|---|---|---|---|
| November *(starting 11-10-2018) | 1,288,400* | 15.2* | 13.9* | 49* | 35* |
| December | 1,773,240 | 12.7 | 7.3 | 32 | 25 |

Studies were conducted throughout the year evaluating BOD concentration at different locations in the disinfection tank. Sampling performed after initiating PAA disinfection showed an increase in soluble BOD downstream of the dosing point compared to the upstream sampling location. This increase in soluble BOD is shown in the above table as well as the calculated Final Effluent BOD adjusting out PAA impact. The increase in soluble BOD appears to be correlated with PAA dose as shown.

Biogas Investigation/Cogeneration

Twelve (12) samples of biogas were collected throughout the year and analyzed for siloxanes, methane, carbon dioxide, oxygen, nitrogen, hydrogen sulfide, and for BTU. Total Siloxanes as silocon averaged 72 ppbv ranging from 11 to 278. Hydrogen Sulfide concentrations averaged

4,043 ppmv ranging from 380 to 8,783. Energy values averaged 591 BTU/cubic feet (wet) ranging from 420 to 660.

Average annual values of analytical results from previous years are summarized as follows:

| Year | Soloxanes (ppbv) | Hydrogen Sulfide (ppmv) | BTUs/cubic foot (wet) |
|---|---|---|---|
| 2012 | 399 | 12,556 | 649 |
| 2013 | 136 | 7,331 | 639 |
| 2014 | 107 | 4,023 | 602 |
| 2015 | NA | 3,129 | 570 |
| 2016 | 159 | 3,266 | 559 |
| 2017 | 131 | 3,083 | 683 |
| 2018 | 72 | 4,043 | 591 |

The Shelby County Health Department granted an extension of the co-gen construction permit through August 25, 2020.

Effluent Discharge Foam Control

The Consent Decree includes the requirement that the City install an engineered structure to reduce air entrainment and turbulence of the Stiles Facility effluent discharge associated with the production of foam. Vortex installation began in September 2015. The major components were in place after mid-2017 and the vortex put into service on 9-20-2017. Before the end of 2017, it became apparent that the foam return line was over-pressurized blowing apart end caps and pipe joints and causing severe erosion down the river bank. Chemical defoamer was added to reduce foam production pending additional modifications. The foam return line was redesigned and a change-order issued. Stiles Facility switched over to the original effluent discharge pipe on 5-4-2018 and the entire foam return line was replaced with heavier duty components including steel piping, risers and heavy concrete thrust blocks around tees. The vortex was put back into operation on 6-11-2018 and the project was inspected and closed out by SRF later in the year. This was by far one of the most successful of the more complex wastewater treatment plant projects undertaken by Public Works.

Laboratory

The laboratory located at the M. C. Stiles Facility received samples from the Stiles Facility, the Maxson Facility, and Chapel Hill package treatment plant. The laboratory performed a total of 25,509 analyses in-house broken down as follows:

| | |
|---|---|
| M.C. Stiles | 13,019 |
| T.E. Maxson | 11,931 |
| Chapel Hill | 559 |
| Industrial monitoring | 0 |
| Total | 25,509 |

Pretreatment Program

The program regulates the discharges of all industrial users, enforces the requirements of the sewer use ordinance, implements the federal categorical standards on specific process wastewaters, and manages the hauled waste program. In 2018, as part of regulating these industrial users, the Control Authority collected 285 samples, performed 107 site inspections, and issued 233 Notices of Violation. There were 104 significant industrial users (SIUs) that were monitored and reported on to the Approving Authority (TDEC).

For the year 2018, the City collected $626,357 for disposal of 4,818,131 gallons of hauled wastewater. Most of the hauled waste deliveries are septic and grease trap related.

The sewer fee history is summarized below:

| Date | Volumetric | BOD Surcharge | TSS Surcharge | Ordinance |
|---|---|---|---|---|
| 5-15-1979 | 28.52 cents | - | - | #2877 |
| 1-1-1982 | 58.68 cents | 2.7 cents | 4.6 cents | - |
| 7-1-2004 | 87.5 cents | 3.65 cents | 5.96 cents | #5036 |
| 7-1-2008 | 98.5 cents | 3.86 cents | 6.21 cents | #5239 |
| 7-1-2009 | 105.3 cents | - | - | #5312 |
| 8-1-2009 | - | 3.88 cents | 6.61 cents | - |
| 7-1-2010 | 226.7 cents | 4.28 cents | 7.16 cents | #5356 |
| 1-1-2018 | 287 cents | - | - | #5651 |

The Maxson Facility rated capacity is 490,000 ppd for influent BOD and 410,000 ppd for influent TSS. The Stiles Facility rated capacity is 275,000 ppd for influent BOD and 374,000 ppd for influent TSS. Loading trends over the past 10 years are shown in Figures 9-11 for the Maxson Facility and in Figures 12-14 for the Stiles Facility. For the Maxson Facility, Figure 9 shows a near perfect alignment in influent BOD and TSS loading after the shutdown of the Cargil fructose operation. Figure 10 shows a steady increase in BOD loading over the past 2 years which may be the result of lagoon recycle. Figure 11 shows increases in TSS loading over the past 4 years that may also be the result of major lagoon recycle. Recycle is also suggested at other times in the past by the relationship between Lagoon Supernatant TSS and influent TSS loading over the years as shown in Figure 12. For the Stiles Facility, Figure 13 shows an increase in TSS loading occurring after KTG increased production. Figure 14 shows a steady correlation between BOD loading and industrial BOD discharge loading over the years. Figure 15 shows increases and changes in TSS loading that are not related to industrial TSS discharge loading (that we know of) and could be related to lagoon recycle and inflow into the collection system during high Mississippi River stages.

Since 1982, the Additional Treatment Cost charged to industries has increased at an annual rate of 1.29 % for BOD and 1.24% for TSS. This low rate of increase represents a very good deal for significant industrial users. The volumetric rate has increased at an annual rate of 6.19 % since 1979.

## 3.0 Analysis and Future Outlook

### Analysis

Table 1 and Table 2 both show influent loadings that have trended with the ups and downs of the economy, industrial expansion and closure, and on-going waste minimization efforts on the part of industrial users. The Maxson Facility treatment capacity history is summarized as follows:

| Year | Flow (mgd) | BOD (ppd) | SS (ppd) |
|---|---|---|---|
| 1975 | 80 | 186,000 | 133,500 |
| 1985 | 80 | 390,000 | 335,000 |
| 1997 | 90 | 490,000 | 410,000 |

The Maxson Facility primary clarifiers (3) were modified during 1991-93 to increase sludge removal for primary dewatering. The 1997 capacity expansion consisted of two ABF Towers, a 4th Primary Clarifier/Thickener, and two Secondary Clarifiers. A covered lagoon capacity expansion project was completed in 2004 which put covered Lagoons 2A, 2B, and 2C into service for improved solids capture and improvements made in primary sludge pumping. Primary sludge dewatering was discontinued in 2004 with primary sludge wasted directly to the lagoons along with waste activated sludge. As a result, approximately 25% of the influent organic loading was diverted from the main plant to the lagoons for co-digestion with waste activated sludge. In 2006, the bio-towers began to be periodically operated in roughing mode for the first time as suggested by Dr. Orris Albertson with great success resulting in energy savings, improved effluent quality, and less waste activated sludge. In 2010, Lagoon No. 1 was recovered and the influent head works replaced. In 2013, the Fine Bar Screens were put into operation and an immediate reduction in plastics and other undesirable items getting into primary and waste activated sludge was noted. This problem that had become more pronounced after the wasting of primary sludge to the lagoons began in 2004. The Primary Scum Removal System was reworked in 2014 increasing the overall removal efficiency of the primary clarifiers.

Co-digestion of primary sludge with waste activated sludge beginning in 2004 lead to a dramatic increase in biogas production that continued through 2016. In 2017, biogas production began falling off. This trend is shown in Figure 16. There has been a decrease in activated sludge wasting in recent years due to apparent less organic loading going to secondary treatment and increased use of roughing mode. This decrease in waste activated sludge wasting increased the ratio of primary to waste activated sludge going to the lagoon. This increased the biogas yield causing the surge in biogas production. However, this surge in biogas production was short-lived. Beginning in 2017, the ratio of primary to waste activated sludge became too rich, causing biogas production to fall off. The only way for this condition to be rectified is to shift more organic loading to secondary treatment by improved primary sludge thickening.

Unmonitored BOD has averaged 181,364 ppd for the last 10 years. Based on population served (an estimated 404,570 people), the expected unmonitored influent BOD should be 125,309 ppd. This is 56,056 ppd greater than expected. Figure 17 compares expected versus actual influent BOD, industrial loading and rated loading capacity. Figure 18 shows unexplained influent BOD versus BOD removed by primary clarifiers. This indicates BOD may be recycling from the lagoons, dewatering, and the fields back to the head of the plant. Strategies to remedy this situation include using Primary Clarifier #4 as a sludge thickener to reduce volume of primary sludge and partitioning Lagoon #5 and wasting into it from multiple points to improve retention. The increase in BOD returning to the head of the plant was totally unexpected after completion of major lagoon improvement projects.

The Stiles Plant has had no major process modifications over the years. Over the past 10 years, unmonitored influent BOD has been stable, averaging 99,679 ppd. The expected unmonitored influent BOD based on population this size (an estimated 427,435 people) is 133,649 ppd.

Since inauguration of the two new lagoon covers in 2007 biogas production has been on a steady decline as shown in Figure 19. During the initial years of operation, as shown in Figure 20, biogas production was directly related to WAS. In more recent years, biogas production was inversely related to WAS. This could be the result of inert solids accumulation and short-circuiting decreasing biogas production. As 2018 was coming to an end, an assessment of covered Lagoon Nos. 1 and 2 was underway to determine the amount of free volume available in the covered lagoons.

Table 3 compares total influent flow, average daily dry weather flow, and average daily billed wastewater flow at both treatment plants compared with drinking water production for each month. The total influent flow is the average daily flow received at the treatment plants. The average daily dry weather flow is based on days with no apparent river or rain impact. The average daily billed wastewater flow is based on residential and commercial water usage and billed large industrial user and wholesale customer sewer usage. Drinking water production is the metered output of the water treatment plants that is sold to residential and commercial customers and a small portion of it is not affiliated with the City. Some industrial users use private wells to generate cooling and process water instead of using MLGW drinking water. Figure 21 shows the Table 3 data over the last 10 years.

Tables 4 and 5 present average annual rain-induced flow per inch of rain, percent of average daily flow that is rain-induced, average daily flow, average daily dry weather flow, annual rainfall, and percent of average daily flow that is river-induced for each treatment facility since 1975 and 1979 for the Maxson and Stiles Facilities, respectively. Trends in rates of rain-induced flow are shown in Figures 22 and 23. Over the last 10 years, rain-induced flow has averaged 28 million gallons per inch of rain at the Maxson Facility and 27 million gallons per inch of rain at the Stiles Facility. Trends in average daily and average daily dry weather flow for both facilities combined are shown in Figures 24. Overall, rain-induced flow has comprised 5% and 4% of average daily flow received at Maxson and Stiles, respectively, and river-induced flow has comprised 1.5% and 9% of average daily flow received, respectively. Combined ADF peaked in 1998 and Combined ADDWF peaked in 1996. Both have been on a steady decline ever since.

Data related to plant operations and sludge production is presented in Tables 6 and 7. Some additional tables presented are Table 8, the monthly plant operating budget expenditures and Table 9, a breakdown of plant expenditures by category. Treatment costs averaged 56 cents per 1000 gallons in 2018, an increase of 4 cents from the previous year as shown in Figure 25 with previous years for comparison. The cost to treat inflow / infiltration was $2,033,124 or 6.4% of expenditures. Tables 10 and 11 comprise the largest industrial user's monthly fees for each plant. Table 12 shows the three wholesale customer monthly fees. These tables show that the fees from these 19 industrial users, hauled wastes, and 3 wholesale customers cover all of the operating cost of the plants, thus allowing the City to maintain low stable rates. Of the $31,716,731 in operating expenditures in 2018, these industrial, hauled waste, and wholesale customers' fees of $40,143,202 amounted to 127% of the costs. The total fees received increased $7,112,889 for the 19 industrial users and hauled waste customers and increased $108,855 for the three wholesale customers from the previous year. All of the revenue increase from the industrial users and hauled waste was due to a sewer rate increase. If it were not for the rate increase, revenue would have decreased $679,586. This additional revenue from industrial users and wholesale customers should off-set new costs related to PAA disinfection in the coming years. Table 13 shows annual hauled waste volume and revenue over the years.

Table 14 is a monthly comparison of the utility charges for both treatment plants. The treatment plants changed to Time of Use contracts effective 10-1-2015 which makes unit cost tracking more complicated due to different on-peak and off-peak time blocks.

Table 15 is a mass balance of the origin of the total suspended solids in the Maxson Facility influent. This table shows that the lagoon supernatant was 10% of nominal plant TSS loading in 2018. Table 16 shows the relationship between total suspended solids in the lagoon supernatant to the pounds of sludge wasted to the lagoon over time. Table 17 shows the total loading of $BOD_5$ of the plant and the amount contributed by the major industries. Table 18 shows the total loading of COD of the plant and the amount contributed by major industries. Table 19 shows the total loading of total suspended solids of the plant and the amount contributed by the major industries. Table 20 shows the water usage and cost. Table 21 is a table showing the electrical usage, cost, biogas production, and propane usage for the covered lagoon. Table 22 is a table showing the electrical usage and cost for other electrical meters. The City uses an average of 2 mgd of potable water at the Maxson Facility for cooling equipment and dewatering spray down and could be asked in the future to transition to a lower quality water source based on future industrial need and potential new regulatory requirements. Table 23 shows long-term trends in BOD loading and wasting.

Table 24 shows the utility usage, cost, and biogas production for the Stiles Facility. Table 25 shows the plant water usage and cost as well as the electrical usage and cost for dewatering. Table 26 shows monthly electrical usage and costs at Chapel Hill. Table 27 shows effluent concentration and compliance history for Chapel Hill. Table 28 shows total loading of $BOD_5$ for the Stiles Facility and the amount contributed to it by the major industries. Table 29 shows the total loading of COD of the plant and the amount contributed by major industries. Table 30 shows the total loading of TSS for the plant and the amount contributed by the major industries. The City uses an average of 1 mgd of potable water at the Stiles Facility for cooling equipment and dewatering spray down and could be asked in the future to transition to a lower quality water source based on future industrial need and potential new regulatory requirements. Biogas purchases by American Yeast are shown in Table 31. Table 32 shows residual solids accumulation rates in Stiles Lagoon Nos. 1, 2, and 3 and disposed sludge cake accumulation rates in the Surface Disposal Site since 1984.

Table 33 summarizes the electrical usage, total cost, unit charges, and savings over the past several years for both treatment facilities. Over the past 10 years, main power charges have increased at an annual rate of 2.56%, main power consumption at an annual rate of 2.07 %, and unit purchase price at an annual rate of 0.55%. Table 34 and Table 35 show comparisons of the Stiles Facility and Maxson Facility influent flows versus the combined flows of the major industries.

Table 36 shows requisitions and purchase orders for goods and services by both plants during 2018 totaling $ $2,135,589. Other goods and services were purchased using negotiated contracts which are not captured in the requisition process.

Final NPDES permits for both treatment plants were issued on 12-2-2011 for the Stiles Facility and on 12-6-2011 for the Maxson Facility, effective 1-1-2012 and expiring on 12-31-2016. Certain elements of the permits were subsequently appealed. TDEC proposed a modification to the Maxson Facility permit in September 2012 seeking to establish a compliance schedule for disinfection keyed to a revised effective permit date. However, the proposed modified permit was never issued. Applications for new permits were submitted 7-5-2016. The City met with TDEC on 4-10-2017 to review the issuance of new permits. TDEC presented information at the meeting indicating the requested secondary treatment adjustments (i.e. variances) would not be granted. On 10-20-2017, the City submitted a report explaining why the variances should be granted as a supplement to the previously submitted permit applications. New draft permits for both treatment plants were public noticed on 11-13-2018. The permits stuck to the nationwide 30-30 standards and did not allow Best Professional Judgement (BPJ) to be used for industry groups without categorical effluent limits for secondary treatment adjustment as was allowed in one particular case when the 2011 permits were issued. Public Works officials drove to Nashville on 12-10-2018 to meet with TDEC and learned that TDEC was not be flexible on this issue. The City was granted an extension request for public comment to 2-15-2019.

Two permitting issues loom on the horizon for both treatment plants. First, a new method demonstrating compliance with 40 CFR 133.102 as it relates to treatment and performance will need to be negotiated with TDEC and agreed upon by the US EPA. At least for the Stiles Facility, the permitting issue will be how to account for the BOD contribution of PAA to final effluent using the existing regulatory compliance point "final effluent" sample location (downstream of the disinfection tank.) For the Maxson Facility, more than likely, secondary treatment will be based on the existing final effluent sample collected upstream of the disinfection tank and PAA impact on BOD would not be considered.

Second, there is the issue of not qualifying for a secondary treatment standard adjustment for effluent limits. Federal secondary treatment standards allow for an effluent BOD or TSS credit in mg/l to be added to the 30-30 limit for industrial categories that exceed 10% of design flow or loading of the POTW. The City has lost large industrial users over the years with categorical treatment standards with numerical limits. These have been replaced with large industrial users in categories without numerical limits. TDEC has been inconsistent in the past about allowing the use of industrial categories with BPJ-derived numerical limits, as discussed below.

Past permit limits are shown in Table 37. The original permits issued in the mid to late 1970s were based on the 30-30 nationwide standards for secondary treatment. Beginning in the late 1970s, the City began requesting a variance from the 30-30 limits based on its large industrial wastewater contribution. In a US EPA memo dated 3-11-1980 from Frank Hall to Sanford Harvey of Region 4, the US EPA stated that variances would not be granted for industrial categories without numerical limits because they had not been public noticed. TDEC subsequently issued permit limits in 1982, 1984, and 1987 allowing the use of Best Professional Judgement (BPJ) for industrial categories without promulgated effluent guidelines because the permit rationale would satisfy the public notice requirement.

Beginning in 1993, TDEC took the opposite stance and disallowed the BPJ approach, stating that the previous permits were issued in error. There was another twist. TDEC included a review of past performance in the permit rationale, which had not been a consideration in previous permits but now played a role as part of anti-backsliding. In 2000, permit limits were issued in the same manner.

In 2011, in the case of the Maxson Facility, TDEC allowed the substitution of Solae in the Grain Mills Industry Category (40 CFR 406) using the BPJ approach. This boosted the secondary treatment adjustment above the previous permit limits and allowed the previous permit limits to continue by default. If this approach had not been taken, the Maxson Facility would have been bound by the 30-30 standards and be in non-compliance upon issuance of these new limits. The Stiles Facility permit limits, on the other hand, did not need to involve BPJ in the secondary treatment adjustment calculations since sufficient qualifying industrial categories with numerical limits were available and were issued only slightly lower than previous permit limits. While the Stiles Facility had a slight reduction in permit concentration limits, it saw the elimination of daily, weekly, and monthly "pounds" as permit limits. This resulted in fewer opportunities to violate compared to the Maxson Facility.

The Maxson Facility Expansion Project was initiated in 2104 at the encouragement of City officials to facilitate expansion of existing industries as well as establishment of new industrial development in the Pidgeon Industrial Park. During the early years of the City of Memphis Pretreatment Program, discharge agreements were signed with industrial users and industries sought to "negotiate" greater discharge limits than currently needed for future expansion. In later years, the City moved to a permit approach which had less perceived opportunities for "negotiation" and began to reign in past promises to industries in an effort to match allotted loading with rated treatment capacity. As such, industries began contacting the Mayor's office in order to secure such commitments outside of the permit.

The quest for expansion began as an $85 million Guaranteed Maximum Price (GMP) proposal from CDM Smith. Standard engineering practice would dictate that the plant expansion would include improved performance to meet 30-30 permit limits on a consistent basis (i.e. $90^{th}$ + percentile level) at maximum design loading conditions. Over time, the expansion project was broken into five smaller separate GMPs which, taken to together, should encompass the desired capacity increase and performance improvements as well as disinfection and influent coarse bar screen replacement. The cost to perform all of this work will be in excess of $100 million once all of the construction packages have been developed with approximately $83 million funding from SRF. The upgrades will go a long way toward ensuring compliance in a worst case regulatory scenario (i.e. 30-30 limits). Any capacity increase, however, associated with the upgrades could potentially cause the City to lose its eligibility for secondary treatment adjustment whether or not BPJ is allowed because the discharge would be below the 10% rated capacity threshold for any one industrial category.

Such a capacity increase and/or performance improvement would need to be implemented at the Stiles Facility for the same reasons (i.e. 30-30 limits) due to the potential to loose eligibility for secondary treatment adjustment. This would be the result of the reduction of flow and loading from existing industries with categorical limits and the current rejection by TDEC of the BPJ approach for industries without promulgated categorical effluent numerical limits. It is a strong possibility that this problem will "take care of itself" once the COD surcharge is implemented on 1-1-2020 and industries begin cutting back their discharge as a result. Other creative measures will need to be implemented such as co-digestion of American Yeast beer waste in the Stiles Facility covered lagoons. Otherwise, primary clarifiers would need to be added to reduce the loading going to secondary treatment.

On 11-27-2007, the Natural Resources Defense Council petitioned the US EPA to request updated information about secondary treatment nutrient removal capability and establish new technology-based nutrient limits as part of the secondary treatment standards. On 3-16-2011, a Nutrient Reduction Framework (NRF) memo was issued to the EPA regional offices to guide EPA technical assistance and collaboration with states. This initiative was subsequently rolled down to the states. On 12-14-2012, the EPA responded to the NRDC petition saying that it agreed the insufficient data exist to draw any general conclusions about the ability of secondary treatment to remove nutrients but that a uniform set of nationally applicable, technology-based nutrient limits was not warranted at this time.

As a result, the EPA initiated a multi-year, multi-phased national study in nutrient discharges from POTWs in order to 1) establish a statistically representative, nationwide baseline for nutrient discharge and removal at these facilities, and 2) characterize operational and management practices that result in improved nutrient reduction. A questionnaire was published in the Federal Register on 9-19-2016 but was withdrawn apparently after the presidential election. As of the end of 2018, no survey questionnaire has been issued to POTWs to complete.

In 2014, TDEC submitted a draft NRF to the EPA for review and subsequently began implementing it. The TDEC NRF prioritizes HUC10 watersheds with 303(d) listed segments. Tennessee is also developing a Non-Point Source nutrient reduction strategy for Mississippi River basin (Hydrogeological Unit IV) using a farmer-led approach so it is not known at this time how the Maxson and Stiles Facilities will be affected. The initial state-wide implementation of nutrient reduction will be an emphasis on optimization instead of capital improvement. In 2018, TDEC's emphasis was on developing a larger stake holder group for input on the NRF in general and developing site-specific criteria for three reservoirs in Tennessee as a priority.

PAA

Continuous PAA dosing began on 11-10-2018 at the Stiles Facility at an average cost of $31,555 per day. This equates to $0.40 per thousand gallons treated based on the trucked rate for PAA and an average dose of 14.1 mg/l. The average effluent color was 812 Pt Co. Based on a long-term ADF of 92 mgd, the average annual cost for PAA at the Stiles Facility is projected to be $11.7 million per year at 46 cents per pound of PAA solution.

The average effluent color for the Maxson Facility measured in 2018 was 421 Pt Co. Based on this effluent color concentration, an average PAA dose of 7.3 mg/l is expected. This equates to $0.18 per thousand gallons treated. Based on a long-term ADF of 70 mgd, the average annual cost for PAA at the Maxson Facility is estimated to be $4.6 million per year.

Performance and Treatment

For the Maxson and Stiles Facilities, the rate of permit compliance for effluent BOD and TSS concentration and percent removal, including results attributable to upsets, bypasses, operational errors, or other unusual conditions, for the past 2 years are as follows:

| | Mason Facility | | Stiles Facility | |
|---|---|---|---|---|
| | BOD Performance | TSS Performance | BOD Performance | TSS Performance |
| Daily | 98.5% | 98.6% | 99.1% | 100% |
| Weekly | 97.2% | 97.2% | 99.7% | 100% |
| Monthly | 91.3% | 83.8% | 100% | 100% |
| Daily % Removal | 100% | 100% | 100% | 99.7% |
| Monthly % Removal | 100% | 100% | 94.9% | 94.4% |

The below table shows what the performance would be if the recently proposed permit limits were applied to the past 2 years:

| | Mason Facility | | Stiles Facility | |
|---|---|---|---|---|
| | BOD Performance | TSS Performance | BOD Performance | TSS Performance |
| Daily | NA | NA | NA | NA |
| Weekly | 77.8% | 65.2% | 85.9% | 84% |
| Monthly | 45.9% | 26.5% | 27% | 44% |

Based on the above tables which include just the federal parameters, the Maxson and Stiles Facility would have 16.6 and 5.9 violations, respectively, based on current permit limits and 44.8 and 33.5 violations, respectively, based on the recently proposed permit limits, during a 12-month period.

Corrosion-The Silent Killer

Coupons were installed at the Maxson Facility from 6-15-2018 to 7-15-2018 and at the Stiles Facility from 5-9-2018 to 6-8-2019 to assess current corrosion rates and project future corrosion conditions. All of the areas surveyed were rated "Severe" with the exception of the Stiles Facility Main Control Room which was rated "Harsh" and the General Lab which was rated "Moderate." The Severe designation means that electronic/electrical equipment is not expected to survive. Each area represents a unique challenge especially as the City transitions to SCADA which involves more delicate hardware with sensitive metals designed for modern control room or data center atmospheres not ~~encountered~~ typically encountered at wastewater treatment plant environments. Results are shown in Tables 38 and 39.

Metals

The Pretreatment Program had set allowable limits for many categorical dischargers that resulted in the significant reduction of the amount of metals received over the past three decades. With the promulgation of the 503 Sludge Regulations, the amount of various metals entering the plants became a greater regulatory concern. Prior to 1994, the discharge of metals had been controlled and regulated but there were no regulations that defined allowable concentrations of various metals in biosolids. Table 40 shows the metals concentrations over the years in the plants' influent and biosolids. The tables utilize the combination of the six most common metals, zinc, copper, nickel, chromium, lead, and cadmium. As the table shows, influent metals loadings have generally decreased over the past 3 decades.

Table 40 shows the average limit that would apply to the Maxson and Stiles Facilities either as land application or surface disposal under the 503 regulations as well as the metals concentrations for biosolids sampled in 2018. The metals are arsenic, cadmium, chromium, copper, lead, mercury, molybdenum, nickel, selenium, and zinc. Metal concentrations in surface-disposed bio-solids have decreased over the past 3 decades at both treatment plants along with influent metals loadings. Metals concentrations may be more of a concern in upcoming years, especially at the Stiles Facility, if dewatered sludge is land applied at agricultural sites with pollutant loading restrictions.

Personnel

Eight new positions for wastewater treatment trainee were filled on 1-24-2018. Two of the trainees entered the program with Grade 3 licenses already. One WTO1 passed the Grade 4 exam and promoted to WTO2 on 5-13-2018. One of the trainees (with Grade 3 license) promoted to maintenance mechanic in mid-year and the vacancy was filled on 10-1-2018 by an existing City employee. As of the end of 2018, none of the trainees have obtained Grade 4 licenses.

Employees of the Month are shown in Table 41. The employee of the month program was discontinued in 2016 in favor of other more general forms of employee recognition. Table 42 shows the yearly number of budgeted positons for each plant over the years.